University of Ottawa

Department of Mathematics and Statistics

MAT 1302B: Mathematical Methods II

Professor: Alistair Savage

Third Midterm Test – White Version

27 March 2015

Surname _____ First Name _____

Student # _____ DGD (1–4) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) Unless otherwise indicated, you must justify your answers to receive full marks.
(d) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this *clearly*. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(e) Write your student number at the top of each page in the space provided.
(f) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(g) You should write in *pen*, not pencil.
(h) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	7	Total
Maximum	4	4	4	5	5	3	3	28
Grade								

QUESTION 1. **[4 points]** For each of the following sets, write 'Y' if the set is a subspace of $\mathbb{R}^n$ for the given n and write 'N' if the set is not a subspace of $\mathbb{R}^n$ for the given n. You will receive 0.5 points for each correct answer, lose 0.5 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

____ The span of 6 vectors in $\mathbb{R}^4$, $n = 4$.

____ The set of solutions to a homogeneous linear system in n variables. Here n can be any positive integer.

____ The set $\{(2x + 3y, 2y - z, x + y + z) \mid x, y, z \in \mathbb{R}\}$, $n = 3$.

____ The set $\{(x, y, x + y, 2) \mid x, y \in \mathbb{R}\}$, $n = 4$.

____ The null space of a 4×5 matrix, $n = 4$.

____ The column space of a 7×8 matrix, $n = 7$.

____ A line in $\mathbb{R}^2$ that does not pass through the origin, $n = 2$.

____ A plane in $\mathbb{R}^3$ containing the origin, $n = 3$.

QUESTION 2. **[4 points]** For each statement below, indicate if it is true (T) or false (F). You will receive 0.5 points for each correct answer, lose 0.5 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

____ The set $\{\vec{0}\}$ is a basis of the subspace $\{\vec{0}\}$.

____ A subspace can have more than one basis.

____ For every $n \times m$ matrix A, the equality $\operatorname{rank} A + \dim \operatorname{Nul} A = m$ holds.

____ If all of the entries of an $n \times n$ matrix A are integers, then the determinant of A is an integer.

____ Every nonconstant polynomial has a complex root.

____ The product of two imaginary numbers is an imaginary number.

____ If V is a subspace of $\mathbb{R}^n$, then every basis of V has n elements.

____ A set of 3 vectors in $\mathbb{R}^5$ is always linearly independent.

QUESTION 3. [**4 points**] Compute the determinant of the matrix

$$A = \begin{bmatrix} -3 & 0 & -2 & 0 & -1 \\ 10 & -2 & 17 & 25 & -18 \\ 5 & 0 & 3 & 0 & 2 \\ -1 & 0 & 2 & 0 & 0 \\ 3 & 0 & -10 & 3 & 8 \end{bmatrix}$$

QUESTION 4. **[5 points]** Find a basis for

$$\text{Span}\left\{\begin{bmatrix}2\\2\\0\\0\\-2\end{bmatrix}, \begin{bmatrix}-4\\-4\\0\\0\\4\end{bmatrix}, \begin{bmatrix}-1\\4\\5\\0\\1\end{bmatrix}, \begin{bmatrix}4\\4\\2\\4\\-4\end{bmatrix}, \begin{bmatrix}3\\5\\10\\16\\-3\end{bmatrix}, \begin{bmatrix}-1\\-3\\-1\\4\\5\end{bmatrix}\right\}$$

What is the dimension of this span?

QUESTION 5. [**5 points**] Find a basis of the null space of the matrix

$$B = \begin{bmatrix} 1 & 1 & 2 & 0 & -1 & -2 \\ 0 & 1 & -2 & -2 & 3 & -5 \\ 0 & 0 & 0 & 1 & -1 & 3 \end{bmatrix}.$$

What is rank B?

QUESTION 6. **[3 points]** Suppose $z = 3 - 2i$ and $w = -3 - 4i$. Calculate

$$\bar{z}w \quad \text{and} \quad \frac{z}{w}.$$

Write your answers in the form $a + bi$, where $a, b \in \mathbb{R}$.

QUESTION 7. Suppose A, B, and C are invertible 3×3 matrices with $\det A = 2$ and $\det C = -2$. Furthermore, suppose that

$$X = 2AB^TCB^{-1}A.$$

(a) **[2 points]** What is $\det X$?

(b) **[1 point]** Is X invertible?

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